

AFFINE CONNECTIONS, CURVATURE, AND TORSION

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1. Principal and linear connections

Definition 1.1. Let M be a smooth manifold and let E be a smooth fiber bundle over M with Ehresmann connection $H \oplus V = TE$. The *curvature* of the connection is the map $\Omega: \Gamma(TE) \times \Gamma(TE) \rightarrow \Gamma(TE)$ defined by

$$\Omega^E(X, Y) = [X_H, Y_H]_V. \quad (1.1)$$

The curvature defines a differential form; indeed, for a smooth function $f \in C^\infty(P)$, we have

$$\Omega^E(X, fY) = [X_H, fY_H]_V = f[X_H, Y_H]_V + [(X_H f)Y_H]_V \quad (1.2)$$

$$= f[X_H, Y_H]_V + (X_H f)[Y_H]_V \quad (1.3)$$

$$= f[X_H, Y_H]_V, \quad (1.4)$$

which is all that is needed to check by antisymmetry and additivity of the Lie bracket. In special cases, one may define a related curvature form on the base manifold M .

1.1. Curvature of principal connections

Given a principal G -bundle $\rho: P \rightarrow M$, define the *adjoint bundle* to be the associated bundle $\text{ad } P = P \times_{\text{Ad}} \mathfrak{g}$, where Ad is the adjoint action of G on $\mathfrak{g}$. Explicitly, we have

$$\text{ad } P = (P \times \mathfrak{g})/\sim; \quad [u, v] \sim [u \cdot g, \text{Ad}_g(v)]. \quad (1.5)$$

We may identify the vertical tangent spaces of P with the fibers of $\text{ad } P$

$$\Psi_u: (\text{ad } P)_{\rho(u)} \rightarrow \ker d\rho_u; \quad [u, v] \mapsto (d\alpha_u)_e(v). \quad (1.6)$$

and $H \subseteq TP$ is a principal Ehresmann connection, then the curvature form Ω^P is

equivariant under the action of G on TP . That is, for all $g \in G$,

$$\Omega^P(g_*X, g_*Y) = [g_*X_H, g_*Y_H]_V = g_*[X_H, Y_H]_V = g_*\Omega^P(X, Y), \quad (1.7)$$

since g_* preserves the decomposition $TP = H \oplus V$ and the Lie bracket. Furthermore, the action $\alpha: P \times G \rightarrow P$ allows us to identify each vertical tangent space $V_u \subseteq T_uP$ with $\mathfrak{g}$ via the isomorphism

$$d\alpha(u, -)_e: \mathfrak{g} \rightarrow V_u \subseteq T_uP. \quad (1.8)$$

Together, these facts allow us to define a curvature form on M by taking horizontal lifts.

Proposition 1.2. *Let $\pi: P \rightarrow M$ be a principal G -bundle with principal Ehresmann connection $H \subseteq TP$. Then H defines a $\mathfrak{g}$ -valued 2-form Ω on M satisfying*

$$d\alpha(u, -)_e(\Omega_p(x, y)) = \Omega_u^P(\tilde{x}_u, \tilde{y}_u) \quad (1.9)$$

for all $u \in \rho^{-1}(p)$.

Proof. The only thing that needs to be shown is that $d\alpha(u, -)_e^{-1}(\Omega_u^P(\tilde{x}_u, \tilde{y}_u))$ does not depend on the choice of u . If $u \in \pi^{-1}(p)$ and $g \in G$, then

$$(d\alpha_{ug})_e = (d\alpha_u)_g \circ (dL_g)_e = \quad (1.10)$$

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1.2. Curvature of linear connections

If $\rho: E \rightarrow M$ is a smooth vector bundle, then for each $e \in E$, there exists an isomorphism

$$\Psi_e: E \rightarrow \ker d\rho_e; \quad f \mapsto \frac{d}{dt} \left[e + tf \right]_{t=0}, \quad (1.11)$$

which may be assembled into a smooth vector bundle isomorphism $\rho^*E \cong \ker d\rho$. Since Ω^E takes values in the vertical subbundle, this defines a map

$$\Omega: \Gamma(TM) \times \Gamma(TM) \times \Gamma(E) \rightarrow \Gamma(E); \quad [\Omega(X, Y)Z]_p = \Omega_{Z_p}^E(\tilde{X}_{Z_p}, \tilde{Y}_{Z_p}), \quad (1.12)$$

where $\tilde{u}_e$ denotes the horizontal lift of a vector $u \in T_pM$ to $e \in \rho^{-1}(u)$. Now, if $H \subseteq TE$ is a linear Ehresmann connection, then

2. Integrating vector-valued forms

Definition 2.1. Let M be a smooth manifold equipped with a smooth null-homotopy $H: M \times I \rightarrow M$ from a point $p \in M$, and let E be a vector bundle over M with linear

connection ∇ . If $\omega \in \Omega^k(M, E)$, then we define

$$\mathcal{P}_H^\nabla \int_M \omega = \int_M \mathcal{P} \quad (2.1)$$

to be the integral of ω over N with respect to H .

Definition 2.2. Let $k > 0$ be an integer. For each integer $0 \leq i < k$ and $\epsilon \in \{0, 1\}$, let $\gamma_i^\epsilon: I \rightarrow I^k$ be a smooth path and $H_i^\epsilon: I^{k-1} \times I \rightarrow I^{k-1}$ be a smooth map such that

1. γ_i^ϵ connects $\mathbf{0} \in I^k$ to some point $y_i^\epsilon = \partial_i^\epsilon \iota_k(x_i^\epsilon)$ in the (i, ϵ) th face, and
2. H_i^ϵ is a homotopy from the constant map at x_i^ϵ to the standard $(k-1)$ -cube ι_{k-1} .

Then the collection $(H_i^\epsilon, \gamma_i^\epsilon)$ is said to be *smooth contraction data* for the k -cube.

Definition 2.3. Let M be a smooth manifold, and let $X = (X_1, \dots, X_k)$ be a tuple of vector fields with global flows $\Phi_{X_i}: M \times I \rightarrow M$. The singular k -cube

$$\pi_X: M \rightarrow \text{Sing}_k^\square(M); \quad x \mapsto [\Phi_{X_1}(x_1) \circ \dots \circ \Phi_{X_k}(x_k)](p) \quad (2.2)$$

is said to be the k -parallelotope spanned by $X_1, \dots, X_k$ based at p .

Theorem 2.4. Let M be a smooth n -manifold and E be a smooth vector bundle over M with linear connection ∇ . Given fixed contraction data $(H_i^\epsilon, \gamma_i^\epsilon)$ for the k -cube, a differential form $\eta \in \Omega^{k-1}(M, E)$, and vector fields $X_1, \dots, X_k$ with unit-time global flow, we have

$$[d^\nabla \eta](X)|_p = \lim_{t \rightarrow 0} \left[\frac{1}{t_1 \dots t_k} \cdot \sum_{i, \epsilon} (-1)^{i+\epsilon} \cdot \mathcal{P}^\nabla(\Pi_p^{tX} \circ \gamma_i^\epsilon) \int_{\partial_i^\epsilon \Pi_p^{tX}} \mathcal{P}^\nabla(\partial_i^\epsilon \Pi_p^{tX} \circ H_i^\epsilon) \eta \right]. \quad (2.3)$$

Proof. Let $U \subseteq M$ be an open neighborhood of p on which $E|_U \cong E_p \times U$, and let ω be the associated connection form in a given trivialization. By continuity, there exists an open neighborhood $W \subseteq \mathbb{R}^k$ of the origin such that $|\Pi_p^{tX}| \subseteq U$ for all $t \in W$. For such t , let

$$J_i^\epsilon(t) = \mathcal{P}^\nabla(\Pi_p^{tX} \circ \gamma_i^\epsilon) \in \text{GL}(E_p); \quad K_i^\epsilon(t) = \int_{\partial_i^\epsilon \Pi_p^{tX}} \mathcal{P}^\nabla(\Pi_p^{tX} \circ H_i^\epsilon) \eta \in E_p, \quad (2.4)$$

which define smooth functions by Proposition ?? and ??. If $t = \tau \mathbf{e}_i$ for some $\tau \in I$, then $\Pi_p^{tX}(x) = \Phi^{X_i}(\tau x_i, p)$. Thus,

$$\Pi_p^{tX} \circ \gamma_i^\epsilon = \Phi_p^{\tau X_i} \circ (\text{proj}_i \circ \gamma_i^\epsilon), \quad (2.5)$$

where $\text{proj}_i: I^k \rightarrow I$ is the projection onto the i th factor. Since $\text{proj}_i \circ \gamma_i^\epsilon$ starts at 0 and ends at ϵ , we have by Proposition ?? and ??,

$$\left. \frac{\partial J_i^\epsilon}{\partial t_i} \right|_{t=0} = \left. \frac{d}{d\tau} \mathcal{P}^\nabla(\Phi_p^{\tau X_i})_0^\epsilon \right|_{\tau=0} = \left. \frac{d}{d\tau} \mathcal{P}^\nabla(\Phi_p^{\epsilon X_i})_0^\epsilon \right|_{\tau=0} = \epsilon \cdot \omega(X_i). \quad (2.6)$$

Now, if $t \in W \cap I^k$, then we may write $\partial_i^\epsilon \Pi_p^{tX} = \sigma_i^{\epsilon t_i} \Pi_p^X \circ \text{diag}(t_{\delta^i(j)})$, where σ is the slice operator defined in (??). Let $\tau \in I$. For all $j < i$, we have

$$\frac{\partial}{\partial x_j} [\sigma_i^{\epsilon \tau} \Pi_p^X]_{x=0} = \frac{\partial}{\partial x_j} [\Phi^{X_i}(\epsilon \tau) \circ \Phi^{X_j}(x_j)] = [\Phi^{X_i}(\epsilon \tau)]_*(X_j), \quad (2.7)$$

and for all $j \geq i$, we have

$$\frac{\partial}{\partial x_j} [\sigma_i^{\epsilon \tau} \Pi_p^X]_{x=0} = \frac{\partial}{\partial x_{j+1}} [\Phi^{X_{j+1}}(x_{j+1}) \circ \Phi^{X_i}(\epsilon \tau)] = [\Phi^{X_i}(\epsilon \tau)]^*(X_{j+1}). \quad (2.8)$$

It follows from Proposition ?? that

$$\begin{aligned} \left. \frac{\partial^{k-1} K_i^\epsilon}{\partial t_1 \cdots \widehat{\partial t_i} \cdots \partial t_k} \right|_{t=\tau \mathbf{e}_i} &= \eta \left([\Phi^{X_i}(\epsilon \tau)]_*(X_1), \dots, [\Phi^{X_i}(\epsilon \tau)]_*(X_{i-1}), \right. \\ &\quad \left. [\Phi^{X_i}(\epsilon \tau)]^*(X_{i+1}), \dots, [\Phi^{X_i}(\epsilon \tau)]^*(X_k) \right) \Big|_{\Phi^{X_i}(\epsilon \tau, p)} \end{aligned} \quad (2.9)$$

and $\partial^\beta K_i^\epsilon|_{t=\tau \mathbf{e}_i} = 0$ for all other multi-indices β of degree less than k . In particular, setting $\tau = 0$ in (??) yields

$$\left. \frac{\partial^{k-1} K_i^\epsilon}{\partial t_1 \cdots \widehat{\partial t_i} \cdots \partial t_k} \right|_{t=0} = \eta(X_1, \dots, \widehat{X_i}, \dots, X_k)|_p \quad (2.10)$$

and the chain rule applied to (??) along the diagonal yields

$$\left. \frac{\partial^k K_i^\epsilon}{\partial t_1 \cdots \partial t_k} \right|_{t=0} = \epsilon \cdot \left[X_i \left(\eta(X_1, \dots, \widehat{X_i}, \dots, X_k) \right) \right. \quad (2.11)$$

$$\left. - \sum_{j < i} \eta(X_1, \dots, X_{j-1}, \mathcal{L}_{X_i} X_j, X_{j+1}, \dots, \widehat{X_i}, \dots, X_k) \right]_p. \quad (2.12)$$

Let $A_i^\epsilon = J_i^\epsilon K_i^\epsilon$ and $A = \sum_{i,\epsilon} (-1)^{i+\epsilon} A_i^\epsilon$. Then for all multi-indices α of degree less than or equal to k , the Leibniz rule yields

$$\partial^\alpha A_i^\epsilon|_{t=0} = \left[J_i^\epsilon \cdot \frac{\partial^k K_i^\epsilon}{\partial t_1 \cdots \partial t_k} + \frac{\partial J_i^\epsilon}{\partial t_i} \cdot \frac{\partial^{k-1} K_i^\epsilon}{\partial t_1 \cdots \widehat{\partial t_i} \cdots \partial t_k} \right]_{t=0}, \quad (2.13)$$

which by (??) and (??) implies $\partial^\alpha A_i^0|_{t=0} = 0$ and

$$\partial^\alpha A_i^1|_{t=0} = \left[X_i \left(\eta(X_1, \dots, \widehat{X_i}, \dots, X_k) \right) + \omega(X_i) \eta(X_1, \dots, \widehat{X_i}, \dots, X_k) \right. \quad (2.14)$$

$$\left. + \sum_{j < i} (-1)^j \eta(\mathcal{L}_{X_i} X_j, X_1, \dots, \widehat{X_j}, \dots, \widehat{X_i}, \dots, X_k) \right]_p$$

$$= \left[\nabla_{X_i} \left(\eta(X_1, \dots, \widehat{X_i}, \dots, X_k) \right) + \sum_{j < i} (-1)^j \eta(\mathcal{L}_{X_i} X_j, X_1, \dots, \widehat{X_j}, \dots, \widehat{X_i}, \dots, X_k) \right]_p. \quad (2.15)$$

Summing (??) over all i with the appropriate sign coefficients, we obtain

$$\frac{\partial^k A}{\partial t_1 \dots \partial t_k} = d^\nabla \omega(X_1, \dots, X_k) \quad (2.16)$$

and $\partial^\alpha A = 0$ whenever $|\alpha| < k$. The conclusion follows from Taylor's theorem. $\blacksquare$

Note that Theorem ?? also implies a separate formula for the exterior covariant derivative in terms of the connection form ω for a trivialization $E|_U \cong E^0 \times U$: by summing (??) over each index, we obtain the identity

$$d^\nabla \eta = d\eta + \omega \wedge \eta \quad (2.17)$$

where $d\eta$ is the exterior derivative of η viewed as an E^0 -valued differential form. Intuitively, this term tracks the *absolute* change in η across opposite faces of the parallelotope in the coordinates of the trivialization, and the term $\omega \wedge \eta$ is a correction which when added yields the change in η across opposite faces discounting the *intrinsic* change in η due to parallel transport. By choosing appropriate contraction data, the preceeding theorem will eventually facilitate the proof of identities involving the exterior covariant derivative through visual, yet rigorous, arguments.

3. Holonomy and curvature

In the 2-dimensional case, there are two useful forms of contraction data that we will make use of, each corresponding to a decomposition of the boundary of I^2 into paths. For

the first type, define

$$x_2^0 = 1 \quad \gamma_2^0 = d_2^0 \quad (3.1)$$

$$x_1^1 = 1 \quad \gamma_1^1 = d_2^0 * d_1^1 \quad (3.2)$$

$$x_2^1 = 0 \quad \gamma_2^1 = d_2^0 * d_1^1 * \overline{d_2^1} \quad (3.3)$$

$$x_1^0 = 0 \quad \gamma_1^0 = d_2^0 * d_1^1 * \overline{d_2^1} * \overline{d_1^0}, \quad (3.4)$$

and let H_i^ϵ be any smooth homotopy from the constant map at x_i^ϵ to the identity. This contraction data corresponds to covering the boundary of a singular square $\pi: I^2 \rightarrow M$ by a single path

$$\partial\pi: I \rightarrow M; \quad \partial\pi = \partial_2^0\pi * \partial_1^1\pi * \overline{\partial_2^1\pi} * \overline{\partial_1^0\pi}. \quad (3.5)$$

in the sense provided by the following proposition.

Proposition 3.1. *Let M be a smooth manifold and E be a smooth vector bundle over M with linear connection ∇ . If $\pi: I^2 \rightarrow M$ is a singular square and $\eta \in \Omega^1(M, E)$, then*

$$\mathcal{P}^\nabla \int_{\partial\pi} \eta = \sum_{i,\epsilon} (-1)^{i+\epsilon} \mathcal{P}^\nabla(\gamma_i^\epsilon) \left[\mathcal{P}_{H_i^\epsilon}^\nabla \int_{\partial_i^\epsilon \pi} \eta \right] \quad (3.6)$$

where $(H_i^\epsilon, \gamma_i^\epsilon)$ is the contraction data defined in (??)–(??).

Proof. Omitted — see Figure ??.

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For the second type of contraction data, define

$$x_2^0 = 1 \quad \gamma_2^0 = d_2^0 \quad (3.7)$$

$$x_1^1 = 1 \quad \gamma_1^1 = d_2^0 * d_1^1 \quad (3.8)$$

$$x_2^1 = 1 \quad \gamma_2^1 = d_1^0 * d_2^1 \quad (3.9)$$

$$x_1^0 = 1 \quad \gamma_1^0 = d_1^0, \quad (3.10)$$

and again let H_i^ϵ be any smooth homotopy from the constant map at x_i^ϵ to the identity. This contraction data corresponds to covering the boundary of a singular square $\pi: I^2 \rightarrow M$ by counterclockwise and clockwise paths

$$\partial^+\pi: I \rightarrow M; \quad \partial^+\pi = \partial_2^0\pi * \partial_1^1\pi, \quad (3.11)$$

$$\partial^-\pi: I \rightarrow M; \quad \partial^-\pi = \partial_2^1\pi * \partial_1^0\pi, \quad (3.12)$$

in the sense provided by the following proposition.

Proposition 3.2. *Let M be a smooth manifold and E be a smooth vector bundle over M*

with linear connection ∇ . If $\pi: I^2 \rightarrow M$ is a singular square and $\eta \in \Omega^1(M, E)$, then

$$\mathcal{P}^\nabla \int_{\partial^+ \pi} \eta - \mathcal{P}^\nabla \int_{\partial^- \pi} \eta = \sum_{i, \epsilon} (-1)^{i+\epsilon} \mathcal{P}^\nabla(\gamma_i^\epsilon) \left[\mathcal{P}_{H_i^\epsilon}^\nabla \int_{\partial_i^\epsilon \pi} \eta \right], \quad (3.13)$$

where $(H_i^\epsilon, \gamma_i^\epsilon)$ is the contraction data defined in (??)–(??).

Proof. Omitted — see Figure ??.

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4. Integrating Lie algebra-valued forms

Let G be a Lie group with Lie algebra $\mathfrak{g}$ and let M be a smooth manifold. If there is a linear connection ∇ on $\mathfrak{g} \times M$, then we may integrate a differential form $\eta \in \Omega^1(M, \mathfrak{g})$ along a curve $\gamma: I \rightarrow M$ to obtain

$$\mathcal{P}^\nabla \int_\gamma \eta = \int_\gamma \left[\mathcal{P}^\nabla(\gamma)_t^0 \right] \eta(\gamma(t)) dt. \quad (4.1)$$

which lies in $\mathfrak{g}$. However, for our purposes, we wish to integrate a path of vectors in $\mathfrak{g}$ and obtain an element of G . To do this, define the *time-ordered exponential* of a function $v: I \rightarrow \mathfrak{g}$, denoted by

$$V(t) = \text{T exp} \int_0^t v(\tau) d\tau, \quad (4.2)$$

to be the unique solution to the differential equation $V'(t) = v(t)V(t)$ with initial value $V(0) = 1$. We may now define the *standard path-ordered integral* of a 1-form $\eta \in \Omega(M, \mathfrak{g})$ along a curve γ to be

$$\text{P exp}^\nabla \int_\gamma \eta = \text{T exp} \int_I [\mathcal{P}^\nabla(\gamma)_t^0] \eta(\gamma(t)) dt. \quad (4.3)$$

This definition generalizes to higher dimensions by iteration: given a function $v: I^k \rightarrow \mathfrak{g}$, define

$$\text{T exp} \int_0^{t_k} \cdots \int_0^{t_1} v(\tau) d^k \tau = \text{T exp} \int_0^{t_k} \left[\text{T exp} \int_0^{t_{k-1}} \cdots \int_0^{t_1} v(\tau) d\tau_1 \cdots d\tau_{k-1} \right] d\tau_k. \quad (4.4)$$

Analogously, we may define the standard path-ordered integral of a k -form $\eta \in \Omega^k(M, \mathfrak{g})$ along a singular k -cube $\pi: I^k \rightarrow M$ by

$$\text{P exp}^\nabla \int_\pi \eta = \text{T exp} \int_{I^k} [\mathcal{P}^\nabla(\gamma_t)] \eta(\pi(t)) d^k t \quad (4.5)$$

where $\gamma_t: I \rightarrow I^k$ is the path

$$\gamma_t = \left(\mathbf{0} \rightarrow t_1 \mathbf{e}_1 \right) * \cdots * \left(\sum_{j < i} t_j \mathbf{e}_j \rightarrow \sum_{j \leq i} t_j \mathbf{e}_j \right) * \cdots * \left(t - t_k \mathbf{e}_k \rightarrow t \right). \quad (4.6)$$

Alternatively, consider the homotopies $H_i: I^k \times I \rightarrow I^k$ given by $H_i(t, \tau) = t - (1 - \tau)t_i \mathbf{e}_i$, and let $H = H_1 * \cdots * H_k$, which is a piecewise-smooth homotopy from the constant map at the origin to the identity. Then we have

$$\mathrm{P} \exp^\nabla \int_\pi \eta = \mathrm{T} \exp \int_{I^k} \left[\mathcal{P}^\nabla(H) \right] \pi^* \eta \quad (4.7)$$

and $\gamma_t = H(t, -)$.

5. Geometric derivations of the Bianchi identities

Theorem 5.1 (The first Bianchi identity). *Let M be a smooth manifold and let E be a smooth vector bundle over M with linear connection ∇ and curvature form Ω . If η is an E -valued 1-form, then*

$$(\mathrm{d}^\nabla)^2 \eta = \Omega \wedge \eta. \quad (5.1)$$

In particular, if η is a differential form valued in a fixed vector space, then $\mathrm{d}^2 \eta = 0$.

Proof. Let $p \in M$ and $X_1, \dots, X_k \in \Gamma(TM)$ be complete vector fields. For $t \in \mathbb{R}^k$, let $\mathcal{P}_t$ be the parallel transport for the pullback bundle $[\Pi_p^{tX}]^* E \cong E_p \times I^k$, and for a differential form ξ , define $\xi_t = [\Pi_p^{tX}]^* \xi$. Given $\epsilon, \delta \in \{0, 1\}$ and integers $1 \leq i \leq k$ and $1 \leq j \leq k-1$, define the paths

$$\alpha_i^\epsilon = \mathbf{0} \rightarrow \epsilon \cdot \mathbf{e}_i; \quad \beta_{i,j}^{\epsilon,\delta} = \epsilon \cdot \mathbf{e}_i \rightarrow \epsilon \cdot \mathbf{e}_i + \delta \cdot \mathbf{e}_{\phi^i(j)}; \quad \gamma_{i,j}^{\epsilon,\delta} = \alpha_i^\epsilon * \beta_{i,j}^{\epsilon,\delta}, \quad (5.2)$$

where $\phi^i: [k-1] \rightarrow [k]$ is the i th face map. Now, fix a homotopy $H: I^{k-2} \times I \rightarrow I^{k-2}$ from the constant map at the origin to the identity, and define

$$A_{i,j}^{\epsilon,\delta}(t) = \mathcal{P}_t(\gamma_{i,j}^{\epsilon,\delta}) \int_{\partial_j^\delta \partial_i^\epsilon \iota_k} \mathcal{P}_t(H) \eta_t; \quad A(t) = \sum_{\substack{i,\epsilon \\ j,\delta}} (-1)^{i+j+\epsilon+\delta} A_{i,j}^{\epsilon,\delta}(t). \quad (5.3)$$

For $j < i$, we have $\partial_j^\delta \circ \partial_i^\epsilon = \partial_{i-1}^\epsilon \circ \partial_j^\delta$, so by Proposition ?? and ??,

$$\begin{aligned}
\left. \frac{\partial^k A}{\partial t_1 \cdots \partial t_k} \right|_{t=0} &= \sum_{\substack{\epsilon, \delta \\ i > j}} (-1)^{i+j+\epsilon+\delta} \left[\frac{\partial^2}{\partial t_i \partial t_j} \left(\mathcal{P}_t(\gamma_{i,j}^{\epsilon,\delta}) - \mathcal{P}_t(\gamma_{j,i-1}^{\epsilon,\delta}) \right) \right. \\
&\quad \cdot \left. \frac{\partial^{k-2}}{\partial t_1 \cdots \widehat{\partial t_j} \cdots \widehat{\partial t_i} \cdots \partial t_k} \int_{\partial_j^\delta \partial_i^\epsilon \iota_k} \mathcal{P}_t(H) \eta_t \right]_{t=0} \\
&= \sum_{\substack{\epsilon, \delta \\ i > j}} \epsilon \delta \cdot (-1)^{i+j} \left[\Omega(X_i, X_j) \eta(X_1, \dots, \widehat{X_j}, \dots, \widehat{X_i}, \dots, X_k) \right]_p \\
&= (\Omega \wedge \eta)(X)|_p.
\end{aligned} \tag{5.4}$$

On the other hand, by Proposition ?? and ??,

$$\begin{aligned}
\left. \frac{\partial^k A}{\partial t_1 \cdots \partial t_k} \right|_{t=0} &= \frac{\partial^k}{\partial t_1 \cdots \partial t_k} \sum_{i,\epsilon} (-1)^{i+\epsilon} \mathcal{P}_t(\alpha_i^\epsilon) \left[\sum_{j,\delta} (-1)^{j+\delta} \mathcal{P}_t(\beta_{i,j}^{\epsilon,\delta}) \int_{\partial_j^\delta \partial_i^\epsilon \iota_k} \mathcal{P}_t(H) \eta_t \right] \Big|_{t=0} \\
&= \frac{\partial^k}{\partial t_1 \cdots \partial t_k} \left[\sum_{i,\epsilon} (-1)^{i+\epsilon} \mathcal{P}_t(\alpha_i^\epsilon) \int_{\partial_i^\epsilon \iota_k} \mathcal{P}_t(H) (d^\nabla \eta)_t \right]_{t=0} \\
&= [(d^\nabla)^2 \eta](X)|_p.
\end{aligned} \tag{5.5}$$

The conclusion follows. ■

The above proof offers some insight into why the second exterior derivative necessarily vanishes, but the second exterior covariant derivative may not: taking the boundary of the boundary of a k -cube results in $(k-2)$ -dimensional “ridges”, and $(d^\nabla)^2$ is the result of integrating η over each ridge. The key observation here is that these come in *pairs* with opposite orientations, but the contribution from each ridge does not immediately cancel out with its pair. Indeed, the integral over each ridge must be parallel transported to the basepoint across two different paths, which form the boundary of a (possibly degenerate) 2-cube. This yields a final contribution equal to the holonomy defect

$$\Omega(\text{square})\eta(\text{ridge}). \tag{5.6}$$

Summing these contributions yields the wedge product $\Omega \wedge \eta$. For the ordinary exterior derivative, no defect exists and the integral over the boundary of the boundary directly vanishes.

The same intuition may also be understood through the usual algebraic derivation of the first Bianchi identity, assuming that we do not take for granted the fact that $d^2 = 0$. Starting from the invariant formula (??) and taking two exterior covariant derivatives, one may interpret each term obtained as one of the expressions seen in above proof. Alternatively,

if the fact that $d^2 = 0$ is already established, then we may directly calculate

$$(d^\nabla)^2 \eta = d^\nabla(d\eta + \omega \wedge \eta) = d^2 \eta + d\omega \wedge \eta - \omega \wedge d\eta + \omega \wedge d\eta + \omega \wedge \omega \wedge \eta = \Omega \wedge \eta, \quad (5.7)$$

which, *prima facie*, has an arguably less clear geometric interpretation.

Theorem 5.2 (The second Bianchi identity). *Let M be a smooth manifold and let E be a smooth vector bundle over M with linear connection ∇ and curvature form Ω . Then*

$$d^\nabla \Omega = 0. \quad (5.8)$$

Proof. We repeat the proof of Theorem ?? up to (??), with ω_t being the connection form on the the pullback bundle $[\Pi_p^{tX}]^* E$. For $1 \leq i \leq 3$ and $1 \leq j \leq 2$, define paths

$$\rho_{i,j}^{\epsilon,\delta} = \epsilon \cdot \mathbf{e}_i + \delta \cdot \mathbf{e}_{\phi^i(j)} \rightarrow \epsilon \cdot \mathbf{e}_i + \delta \cdot \mathbf{e}_{\phi^i(j)} + \mathbf{e}_{\phi^i(\phi^j(1))} \quad (5.9)$$

parameterizing the “ridges” $\partial_j^\delta \partial_i^\epsilon \iota_3$. Then for $j < i$,

$$A_{i,j}^{\epsilon,\delta}(t) - A_{j,i-1}^{\epsilon,\delta}(t) = \mathcal{P}_t(\rho_{i,j}^{\epsilon,\delta} * \gamma_{i,j}^{\epsilon,\delta}) - \mathcal{P}_t(\rho_{j,i-1}^{\epsilon,\delta} * \gamma_{j,i-1}^{\epsilon,\delta}), \quad (5.10)$$

and $\rho_{i,j}^{\epsilon,\delta} = \rho_{j,i-1}^{\epsilon,\delta}$. To compute the derivative, we first “close” the loop, applying the Leibniz rule to obtain

$$\begin{aligned} \frac{\partial^3}{\partial t_1 \partial t_2 \partial t_3} [A_{i,j}^{\epsilon,\delta}(t) - A_{j,i-1}^{\epsilon,\delta}(t)]_{t=0} &= \frac{\partial^3}{\partial t_1 \partial t_2 \partial t_3} \left[\mathcal{P}_t \left(\overline{\rho_{j,i-1}^{\epsilon,\delta} * \gamma_{j,i-1}^{\epsilon,\delta}} \right) \left(A_{i,j}^{\epsilon,\delta}(t) - A_{j,i-1}^{\epsilon,\delta}(t) \right) \right]_{t=0} \\ &= \frac{\partial^3}{\partial t_1 \partial t_2 \partial t_3} \left[\mathcal{P}_t \left(\overline{\gamma_{j,i-1}^{\epsilon,\delta}} * \overline{\rho_{j,i-1}^{\epsilon,\delta}} * \rho_{i,j}^{\epsilon,\delta} * \gamma_{i,j}^{\epsilon,\delta} \right) - \mathbf{1} \right]_{t=0} \\ &= \frac{\partial^3}{\partial t_1 \partial t_2 \partial t_3} \left[\mathcal{P}_t \left(\overline{\gamma_{j,i-1}^{\epsilon,\delta}} * \gamma_{i,j}^{\epsilon,\delta} \right) - \mathbf{1} \right]_{t=0} \\ &= 0, \end{aligned} \quad (5.11)$$

since the final term only depends on 2 coordinates. It follows from Theorem ?? and (??) that $[d^\nabla \Omega](X)|_p = (\Omega \wedge \omega)(X)|_p = 0$. $\blacksquare$

The above proof demonstrates that the second Bianchi identity may be understood as a special case of the first Bianchi identity for the connection form ω . To conclude, we give a brief explanation of the geometric picture behind the argument. Evaluating ω over a ridge gives the change due to parallel transport across the ridge. Applying the holonomy defect around the attached square, we find that these contributions do cancel out in pairs, as they form “degenerate” loops that may be pinched off. As a result, we have

$$\Omega(\text{square})\omega(\text{ridge}) = 0. \quad (5.12)$$

It follows that $\Omega \wedge \omega = 0$, so the exterior covariant derivative of the curvature form vanishes.